

Improve Your Sensor Precision with Auto-Zero Operational Amplifiers

Microcontrollers

Digital Signal
Controllers

Analog

Serial
EEPROMs



With their low operating-voltage range of 1.8V to 5.5V and low typical quiescent current of just 300 μ A, the **MCP6V0X op amps** from Microchip Technology can run off of two 1.5V cells with full battery utilization. The devices feature a self-correcting architecture that results in an extremely low maximum offset voltage of 2 μ V and minimal drift over time and temperature. Additionally, their rail-to-rail input/output structure enables greater dynamic range, resulting in better system performance at all operating voltages.

Technical Features:

- Low Offset Voltage, 2 μ V maximum
- Low Voltage Operation, 1.8V to 5.5V
- Low Quiescent Current: 300 μ A typical for GBWP of 1.3 MHz
- Rail-to-Rail Input and Output
- Common Mode Rejection Ratio of 130 dB minimum
- Power Supply Rejection Ratio of 130 dB minimum
- Specified Over an Extended Temperature Range of -40°C to +125°C

The **MCP6V01 Reference Design Board** (Part #: MCP6V01RD-TCLP) demonstrates how to use a difference amplifier system to measure EMF voltage at the cold junction of a thermocouple in order to accurately measure the temperature of the thermocouple bead. This can be done using the MCP6V01 auto-zero op amp because of its ultra-low offset voltage and high Common Mode Rejection Ratio.



Bangalore 91-80-30904444 • New Delhi 91-11-41608632 • Pune 91-20-25661512 • Email Inquiry: krita.nelapudi@microchip.com

www.microchip.com/analog

